

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.1: Data Representation

Concept 1.1.5: Binary Arithmetic (Addition, Subtraction, Overflow)

Total Questions: 18

Mark Schemes begin on Page 9

Question 1 | Source: 9618_w25_qp_11 (Q1.b.i)

(b) (i) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \\ +\ 0\ 1\ 1\ 1\ 1\ 0\ 0\ 0 \\ \hline \end{array}$$

[1]

Question 2 | Source: 9618_w25_qp_11 (Q1.b.ii)

(ii) Name and describe the error that can occur when binary addition is performed.

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.....

.....

..... [2]

Question 3 | Source: 9618_w25_qp_12 (Q7.a)

7 (a) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 0\ 0\ 0\ 1\ 1\ 1 \\ +\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 1 \\ \hline \end{array}$$

[1]

Question 4 | Source: 9618_s25_qp_13 (Q7.b)

(b) Complete the following binary addition. Show your working. Include any overflow bit(s).

$$\begin{array}{r} 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\ + \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \\ \hline \end{array}$$

[1]

Question 5 | Source: 9618_s25_qp_13 (Q7.c)

(c) Subtract the binary number 00011110 from the binary number 01100100 using binary subtraction. Show your working.

[2]

Question 6 | Source: 9618_w24_qp_11 (Q1.c)

(c) Subtract the denary number 23 from the two's complement binary number 01001010

Perform this calculation using binary subtraction.

Show your working.

Working

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.....

.....

.....

Answer

[2]

Question 7 | Source: 9618_w24_qp_11 (Q1.d)

(d) State **one** reason why binary addition and subtraction can result in overflow.

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..... [1]

Question 8 | Source: 9618_s24_qp_11 (Q7)

7 Complete the binary addition. Show your working.

$$\begin{array}{r} 1\ 0\ 0\ 1\ 1\ 1\ 1\ 0 \\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 1 \\ +\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 1 \\ \hline \end{array}$$

[3]

Question 9 | Source: 9618_s24_qp_12 (Q7.b)

7 A computer stores binary data.

- (b) Subtract the denary number 10 from the denary number 100 using binary subtraction.

Show your working.

Working

.....

.....

.....

.....

.....

Answer

[3]

Question 10 | Source: 9618_s24_qp_13 (Q1.c)

- (c) State what is meant by an **overflow in binary addition**.

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..... [1]

Question 11 | Source: 9618_s23_qp_11 (Q3.d.v)

- 3 A computer has an Operating System (OS).

- (d) The computer stores data in binary form.

- (v) Add the following two binary integers using binary addition. Show your working.

$$\begin{array}{r} 10110000 \\ + 00011011 \\ \hline \end{array}$$

[2]

Question 12 | Source: 9618_w22_qp_11 (Q1.b)

(b) Perform the following binary subtraction. Show your working.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \\ -\ 0\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\ \hline \end{array}$$

[2]

Question 13 | Source: 9618_w22_qp_12 (Q2.b)

(b) Perform the following binary addition.

$$\begin{array}{r} 1\ 0\ 0\ 0\ 1\ 1\ 0\ 0 \\ +\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \\ \hline \end{array}$$

[1]

Question 14 | Source: 9618_w21_qp_11 (Q1.b.i)

(b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ +\ 00110111 \\ \hline \end{array}$$

[2]

Question 15 | Source: 9618_w21_qp_11 (Q1.b.ii)

(ii) State how an overflow can occur when adding two binary integers.

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..... [1]

Question 16 | Source: 9618_w21_qp_12 (Q4.e.i)

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---

(e) The binary contents of **two** registers are:

Register 1	0	0	1	1	1	1	0	1
Register 2	0	0	1	0	1	1	0	1

(i) Add the contents of **Register 1** and **Register 2**. Show your working.

Answer [2]

Question 17 | Source: 9618_w21_qp_12 (Q4.e.ii)

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---

(e) The binary contents of **two** registers are:

Register 1	0	0	1	1	1	1	0	1
Register 2	0	0	1	0	1	1	0	1

(ii) Subtract the contents of **Register 2** from the contents of **Register 1**. Show your working.

Answer [2]

Question 18 | Source: 9618_s21_qp_11 (Q1.c.ii)

1 Anya scans an image into her computer for a school project.

(c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

- (ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

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.....

.....

Answer (in binary)

[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q1.b.i)

1(b)(i)	(1) 00101011	1
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Mark Scheme for Question 2 | Source: 9618_w25_ms_11 (Q1.b.ii)

1(b)(ii)	1 mark per bullet point, max 2 marks - An overflow error occurs - The answer cannot be represented in the number of bits available // The answer is larger than the maximum positive number that can be stored in the register // The answer is smaller than the most negative number that can be stored in the register	2
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Mark Scheme for Question 3 | Source: 9618_w25_ms_12 (Q7.a)

7(a)	1 mark for the correct answer 1100 0000	1
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Mark Scheme for Question 4 | Source: 9618_s25_ms_13 (Q7.b)

7(b)	1 mark for $\begin{array}{r} 1\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\ +_1 1\ 1\ 0\ 1\ 1\ 1\ 0\ 0\ 0\ 1 \\ \hline (1) 1\ 0\ 1\ 0\ 0\ 1\ 1\ 0 \end{array}$ Answer: (1) 1010 0110	1
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Mark Scheme for Question 5 | Source: 9618_s25_ms_13 (Q7.c)

7(c)	1 mark for showing binary subtraction (any method) Direct subtraction: $\begin{array}{r} 0\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 0 \\ -0\ 0\ 0\ 1\ 1\ 1\ 1\ 1\ 0 \\ \hline 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array}$ Adding the two's complement: $\begin{array}{r} 0\ 1\ 1\ 0\ 0\ 1\ 0\ 0 \\ +_1 1\ 1\ 1\ 1\ 0\ 0\ 0\ 1\ 0 \\ \hline (1) 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array}$ 1 mark for answer 0100 0110	2
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Mark Scheme for Question 6 | Source: 9618_w24_ms_11 (Q1.c)

1(c)	1 mark for working 1 mark for answer. Working Using the two's complement and adding: 23 in binary = 00010111 so minus 23 is 11101001 $\begin{array}{r} 0100\ 1010 \\ +\ 1110\ 1001 \\ \hline (1)0011\ 0011 \end{array}$ Direct subtraction: $\begin{array}{r} 0100\ 1010 \\ -\ 0001\ 0111 \\ \hline 0011\ 0011 \end{array}$ Answer: 0011 0011	2
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Mark Scheme for Question 7 | Source: 9618_w24_ms_11 (Q1.d)

1(d)	1 mark from: - The answer cannot be represented in the number of bits available - The answer is larger than the maximum positive number that can be stored in the register - The answer is smaller than the most negative number that can be stored in the register	1
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Mark Scheme for Question 8 | Source: 9618_s24_ms_11 (Q7)

7	1 mark each: - Working – carried values clearly indicated - Correct answer 0001 1000 - Overflow clearly indicated as overflow Example: $\begin{array}{r} 1\ 0\ 0\ 1\ 1\ 1\ 1\ 0 \\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 1 \\ +\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 1 \\ \hline (1)\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 0 \\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1\ldots\ldots\ldots(\text{carries}) \end{array}$	3
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Mark Scheme for Question 9 | Source: 9618_s24_ms_12 (Q7.b)

7(b)	1 mark each: - Converting 100 to binary 0110 0100 and 10 to binary 0000 1010 - Subtraction method - converting 10 to – 10 and adding // direct subtraction correct answer 0101 1010 Method 1: Converting to -10 and adding: Binary for +10 is 0000 1010 Binary for –10 is 1111 0110 Binary for 100 is 0110 0100 100 + (-10): <pre> 0 1 1 0 0 1 0 0 +1 1 1 1 0 1 1 0 (1) 0 1 0 1 1 0 1 0 Carries: 1 1 0 0 1 0 0 0 </pre> Method 2: Direct Subtraction Borrows: <pre> 0 0 0 1 1 0 1 0 0 1 1 0 0 1 0 0 - 0 0 0 0 1 0 1 0 0 1 0 1 1 0 1 0 </pre>	3
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Mark Scheme for Question 10 | Source: 9618_s24_ms_13 (Q1.c)

1(c)	1 mark for a correct answer: The answer is too long to be represented in the same number of bits as the binary numbers being added	1
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Mark Scheme for Question 11 | Source: 9618_s23_ms_11 (Q3.d.v)

3(d)(v)	1 mark for working 1 mark for answer <pre> 1 0 1 1 0 0 0 0 + 0 0 0 1 1 0 1 1 1 1 0 0 1 0 1 1 1 1 </pre>	2
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Mark Scheme for Question 12 | Source: 9618_w22_ms_11 (Q1.b)

1(b)	1 mark for working, 1 mark for answer (0011 1110) Working using borrowing: <pre> 1 10 10 1 0 10 0 0 10 10 1 0 1 1 0 0 1 1 0 1 1 1 0 1 0 1 - ----- 0 0 1 1 1 1 1 0 </pre> Working using two's complement: <pre> 1 0 1 1 0 0 1 1 Two's complement 1 0 0 0 1 0 1 1 + ----- (1) 0 0 1 1 1 1 1 0 1 1 </pre>	2
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Mark Scheme for Question 13 | Source: 9618_w22_ms_12 (Q2.b)

2(b)	1101 0010	1
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Mark Scheme for Question 14 | Source: 9618_w21_ms_11 (Q1.b.i)

1(b)(i)	1 mark for answer 1 mark for working e.g. <pre> 1010 1010 0011 0111 <u>1110 0001</u> 1 1 1 1 1 1 </pre>	2
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Mark Scheme for Question 15 | Source: 9618_w21_ms_11 (Q1.b.ii)

1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
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Mark Scheme for Question 16 | Source: 9618_w21_ms_12 (Q4.e.i)

4(e)(i)	1 mark for working, 1 mark for answer $\begin{array}{r} 0011 \ 1101 \\ +0010 \ 1101 \\ \hline 0110 \ 1010 \\ 111 \ 1 \ 1 \end{array}$	2
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Mark Scheme for Question 17 | Source: 9618_w21_ms_12 (Q4.e.ii)

4(e)(ii)	1 mark for working, 1 mark for answer $\begin{array}{r} 0011 \ 1101 \\ +1101 \ 0011 \text{ (two's complement)} \\ \hline 0001 \ 0000 \\ 1111 \ 111 \end{array}$	2
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Mark Scheme for Question 18 | Source: 9618_s21_ms_11 (Q1.c.ii)

1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer $\begin{array}{r} 0010 \ 0011 \\ + \ 0000 \ 1111 \\ \hline \end{array}$	3
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